114 Fall Programing Design

"Basic Billing & Large Order Flag"

Problem description

Yummy Catering charges $35 per guest. Write a Java program that reads a single integer N from standard input, representing the number of guests attending an event. The program must display the number of guests, the fixed price per guest, the total price, and whether the event is a large event. An event is considered large when the number of guests is greater than or equal to 50.

Constraints

- Input: a single integer N, where 0 ≤ N ≤ 10000.
- Use standard input/output only (System.in / System.out).
- Price per guest is fixed at $35.
- Output exactly four lines in the required format.
- Large event is true when N ≥ 50; otherwise it is false.

Sample input / output

Sample input and output for this problem:

Input	Output
30	Guests: 30 Price per guest: $35 Total: $1050 Large event: false
55	Guests: 55 Price per guest: $35 Total: $1925 Large event: true

114 Fall Programing Design

"Rental Breakdown with Modular Methods"

Problem description

Sunshine Seashore Supplies charges $40 per full hour of rental and $1 for each additional minute. Write a Java program that reads the total rental time in minutes and uses separate static methods to display the company motto, calculate full hours, calculate remaining minutes, and calculate the total price.

Your program must include the following:

- displayMotto(): prints the three-line motto and border.
- computeHours(int totalMinutes): returns the number of complete hours.
- computeExtraMinutes(int totalMinutes): returns the remaining minutes.
- computePrice(int hours, int extraMinutes): returns the total rental price.

Constraints

- Input: a single integer M, where 0 ≤ M ≤ 100000.
- Use standard input/output only (System.in / System.out).
- A full hour costs $40 and each extra minute costs $1.
- The output must begin with the three-line motto shown in the sample.
- After the motto, output Hours, Minutes, and Total on separate lines.

Sample input / output

Sample input and output for this problem:

Input	Output
135	SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS S Sunshine Seashore makes it fun in the sun. S SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS Hours: 2 Minutes: 15 Total: $95

114 Fall Programing Design

"Event Object & Encapsulation"

Problem description

Write a Java program that defines an Event class to store information about a Yummy Catering event. The program reads an event number and the number of guests, creates an Event object, and displays the event information. The price is $35 per guest, and an event is considered large when it has at least 50 guests.

Your program must include the following:

- Event(String eventNumber, int guests): constructor that stores the event number and sets the number of guests.
- setGuests(int guests): stores the guest count and updates the total price.
- getEventNumber(), getGuests(), and getPrice(): return the corresponding field values.
- isLargeEvent(): returns true when the number of guests is at least 50; otherwise returns false.

Constraints

- Input: one String event number and one integer number of guests, separated by a space.
- Use standard input/output only (System.in / System.out).
- Price per guest is fixed at $35.
- The Event fields eventNumber, guests, and price must be private.
- Output exactly four lines in the required format.

Sample input / output

Sample input and output for this problem:

Input	Output
A101 60	Event Number: A101 Guests: 60 Total Price: $2100 Is Large: true

114 Fall Programing Design

"Fair Rental Price & Equipment Selection"

Problem description

Sunshine Seashore Supplies offers three types of rental equipment. Write a Java program that reads an equipment type code and the total rental time in minutes. The rental fee is calculated according to the following rules:

- **Rule 1 – Equipment type:** Type 1 is Kayak, Type 2 is Canoe, and Type 3 is Paddleboard.
- **Rule 2 – Hourly rate:** Kayak costs $40 per hour, Canoe costs $45 per hour, and Paddleboard costs $50 per hour.
- **Rule 3 – Additional minutes:** Extra minutes normally cost $1 per minute. If the number of extra minutes is greater than 40, the additional charge is the full hourly rate for the selected equipment.
- **Rule 4 – Invalid input:** If the equipment type is not 1, 2, or 3, or if the total rental time is negative, output Invalid Input and terminate the program.

Constraints

- **Input:** two integers separated by a space: equipment type and total rental minutes.
- Use standard input/output only (System.in / System.out).
- For valid input, output Equipment, Hours, Extra Minutes, and Total on separate lines.

Sample input / output

Sample input and output for this problem:

Input	Output
1 80	Equipment: Kayak Hours: 1 Extra Minutes: 20 Total: $60
1 110	Equipment: Kayak Hours: 1 Extra Minutes: 50 Total: $80
4 60	Invalid Input